

Case Study: Cyclistic Bike-Share Analysis

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Tools used: SQL(BigQuery), Google DOCS,
GoogleSheets, Tableau

1. Executive Summary and Business Task (Ask) :

Context:

Cyclistic is a successful bike-share program in Chicago. Financial analysis reveals that annual members are significantly more profitable than casual riders. Based on this insight, the marketing director aims to design marketing strategies aimed at converting casual riders into annual members.

Key Business Task:

Identify key behavioral differences between casual riders and annual members using 12-month trip data from 2025. If you stop reading here, the last thing I ask you is to go all the way down and read key takeaways.

Core Analysis Questions:

- 1) How do annual members and casual riders use Cyclistic bikes differently?
- 2) Why would casual riders buy a Cyclistic annual membership?
- 3) How can Cyclistic use digital media to influence casual riders to become members.

2. Data Description and Preparation (Prepare):

Data Source:

The analysis utilizes 12 monthly datasets of Cyclistic trip data for the year 2025. The dataset includes:

Unique trip identifiers (ride_id)

Bike type (rideable_type: classic, electric)

Trip start and end timestamps (started_at, ended_at)

User category (member_casual: member, casual)

Data Privacy:

The dataset has been anonymized. Personally identifiable information (PII), such as credit card numbers or individual user profiles, is excluded in compliance with data privacy standards.

3. Data Cleaning and Transformation (Process):

The 12 individual monthly tables were transformed and consolidated into a unified master dataset within Google BigQuery through the following steps:

Aggregation: Merged 12 monthly tables using the UNION ALL operation.

Feature Engineering: Calculated ride_length in seconds via `TIMESTAMP_DIFF(ended_at, started_at, SECOND)` and extracted day of the week

Data Cleaning & Anomaly Removal:

Removed negative duration records caused by system clock desynchronization.

Filtered out trips ≤ 180 seconds to eliminate false starts, lock testing, and unridden bikes caused by technical faults.

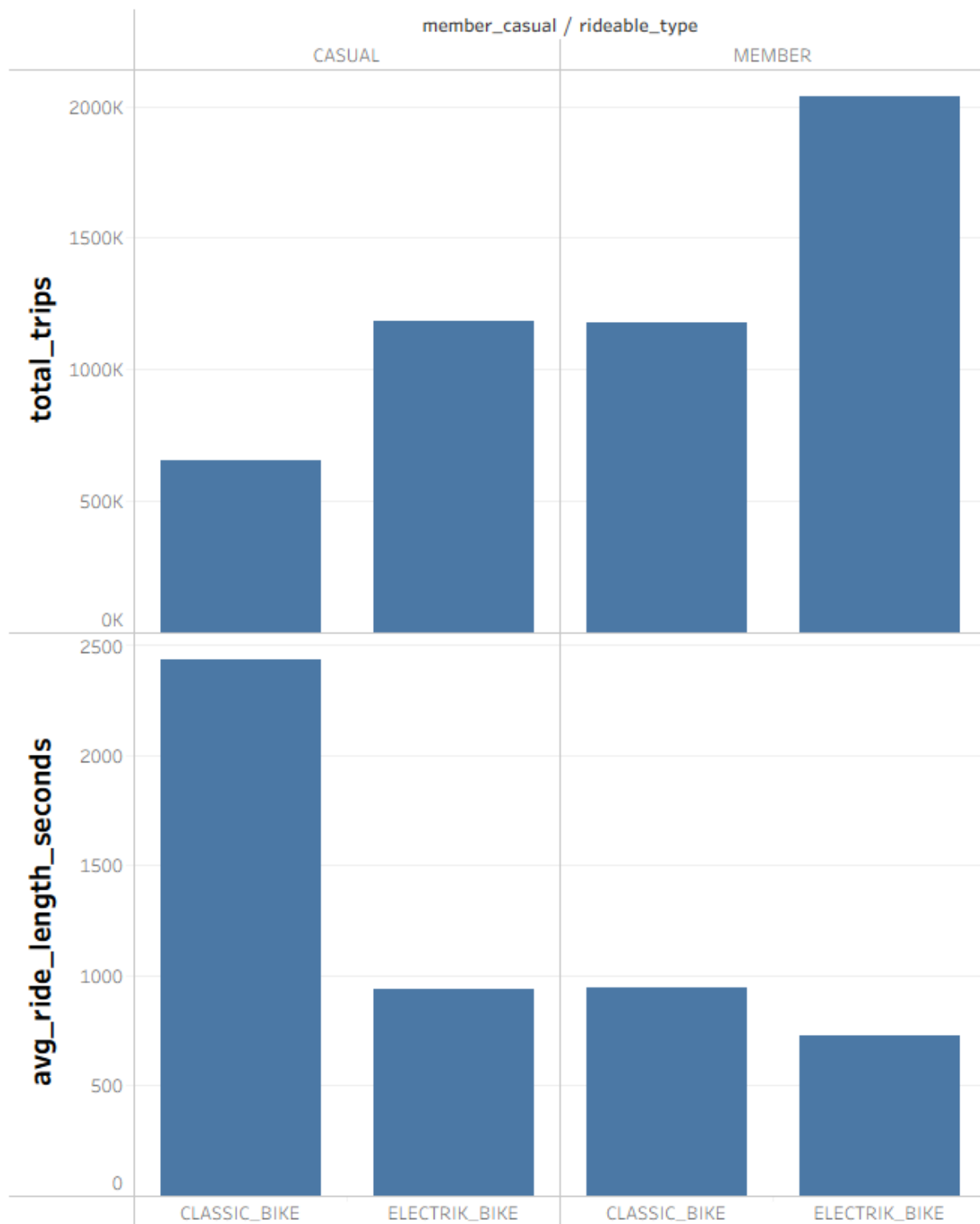
4. Exploratory Data Analysis (Analyze):

Row	member_casual ▼	total_trips ▼	avg_ride_length_...
1	member	3214168	805.6970690393...
2	casual	1836165	1468.789681755...

While casuals have **82%** longer average ride_length, it's noticeable that members have **57%** more of total trips.

Row	member_casual	rideable_type	total_trips	avg_ride_length_...
1	member	classic_bike	1176372	943.3276268051...
2	member	electric_bike	2037796	726.2461649743...
3	casual	electric_bike	1182468	935.4946653947...
4	casual	classic_bike	653697	2433.463362995...

For a more clear view of what happens in this table, I transferred it into Tableau graphs



Looking on the first bar graph, overall, the number of rides taken by members using classic and electric bikes are **79.9%** and **72.3% higher** than those taken by casual riders, respectively.

Firstly, it can be seen that casuals prefer electric bikes over classic ones and the amount of rides taken with electrical vehicles is almost twice as much as usual bicycles.

Likewise, members are more likely to choose electric bikes on a daily basis as the number of rides taken with those is almost twice as much as rides with classic bikes.

By calculating the percentage of electric/classical bikes, I got **80%** and **73% higher** preference for electric bikes by casuals and members respectively.

Taking a look at the second chart, there is eventually a noticeable difference between the average time for a ride casuals have on classic bicycles and on electric ones, however, there is no similar contrast in members' average ride length time.

The average length for a ride that casual has on a classic bike is **2.6** times the length of a ride on an electric one.

While members use usual bicycles on the average for no more than just **1.3** times.

Row	member_casual	rideable_type	day_of_week	total_trips	avg_ride_length...
1	casual	classic_bike	1	119670	2655.377220690...
2	casual	classic_bike	2	75094	2399.262843902...
3	casual	classic_bike	3	69802	2212.612933726...
4	casual	classic_bike	4	64968	2113.684891023...
5	casual	classic_bike	5	75923	2248.804881261...
6	casual	classic_bike	6	100117	2492.457634567...
7	casual	classic_bike	7	148123	2570.622941744...
8	casual	electric_bike	1	185453	1081.037297859...
9	casual	electric_bike	2	134217	923.5682737656...
10	casual	electric_bike	3	137435	819.9110052024...
11	casual	electric_bike	4	137700	796.9595860566...
12	casual	electric_bike	5	160878	836.1160071607...
13	casual	electric_bike	6	193558	932.2218404819...
14	casual	electric_bike	7	233227	1047.798338099...
15	member	classic_bike	1	132262	1049.200488424...
16	member	classic_bike	2	173800	903.4487629459...
17	member	classic_bike	3	189931	909.4471044747...
18	member	classic_bike	4	179613	899.0585870733...
19	member	classic_bike	5	184877	905.2609951481...
20	member	classic_bike	6	167586	944.3757533445...
21	member	classic_bike	7	148303	1038.917331409...
22	member	electric_bike	1	213481	795.9257498325...
23	member	electric_bike	2	279193	700.7225682592...
24	member	electric_bike	3	320710	701.2580430918...
25	member	electric_bike	4	318361	695.1970655953...
26	member	electric_bike	5	336594	705.2381801220...
27	member	electric_bike	6	310050	726.9939493630...
28	member	electric_bike	7	259407	791.7371350811...

By watching on the table, it firstly looks like a ton of different and non understandable integers, but it is almost the same we have encountered before. Let me explain it to you my valuable stakeholder. Days of the week in my SQL table are ordered as 1=Sunday and 7=Saturday.

In this table we have everything the same as before but grouped by days of the week to clearly see patterns based on weekdays

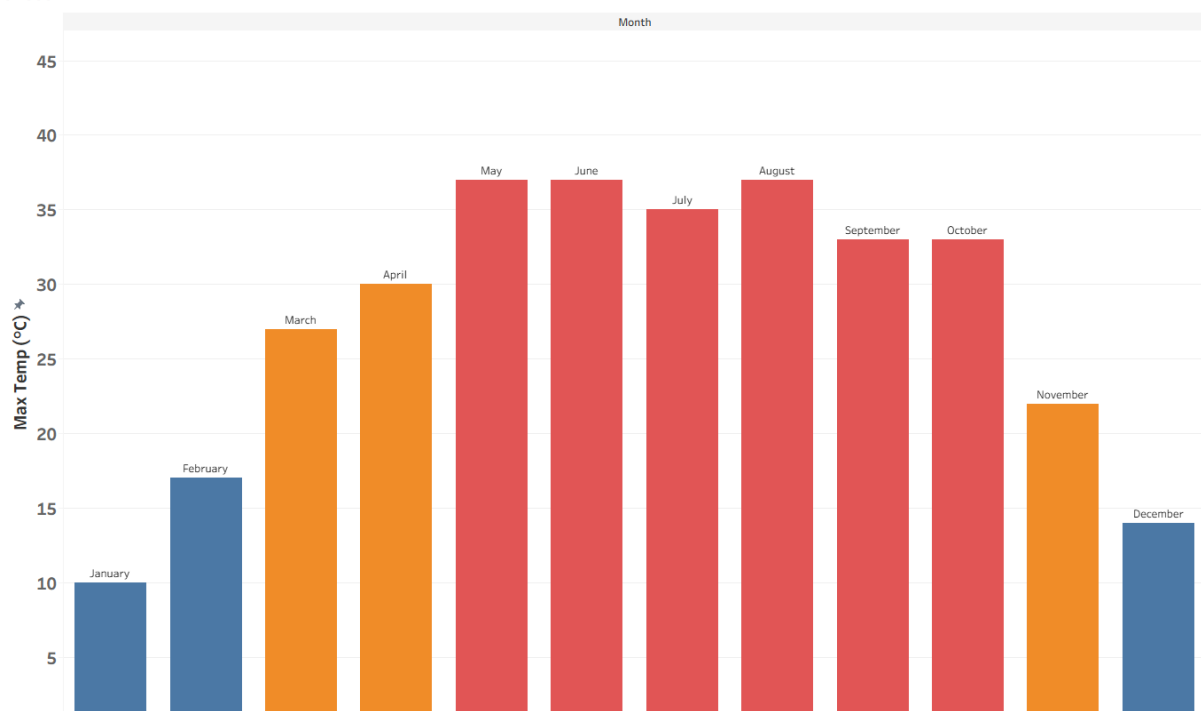
Day of Week Usage Dynamics:

Overall, it is clear that casual riders use bikes mostly on weekends and prefer electric ones, while members show the opposite trend by riding mainly on weekdays.

Looking at the numbers, casual users take between **65 k** and **76 k** classic bike rides on workdays with an average duration of **2,1 k** to **2,4 k** seconds, but this amount almost doubles on weekends, reaching a peak of **148.123** rides. In contrast, members have lower numbers on weekends and way higher on working days.

The next thing I want to do is to analyze different correlations depending on the overall temperature. Before doing all these, I actually have a theory that the colder the temperature, the less people use bicycles. For this task, I searched in the browser to find the archive with the temperature of the USA, Chicago in all 12 months. Here is the graph result after transferring it into Tableau bar chart:

Sheet 1



Blue means cold month, orange means mild temperature in month and red means hot. Based on this, I will present tables in SQL and show conclusions based on temperature contribution to the Cyclistic company.

```
SELECT
    member_casual,
    rideable_type,
    started_at,
    ended_at,
    TIMESTAMP_DIFF(ended_at, started_at, SECOND) AS ride_length,
    AVG(TIMESTAMP_DIFF(ended_at, started_at, SECOND)) AS avg_ride_length
FROM `Capstone_Project_Cyclistic.January_2025`
WHERE TIMESTAMP_DIFF(ended_at, started_at, SECOND) > 180
GROUP BY
    member_casual,
    rideable_type,
    started_at,
    ended_at
UNION ALL
```

This code was used as a **function** for creating multiple temperature based tables.

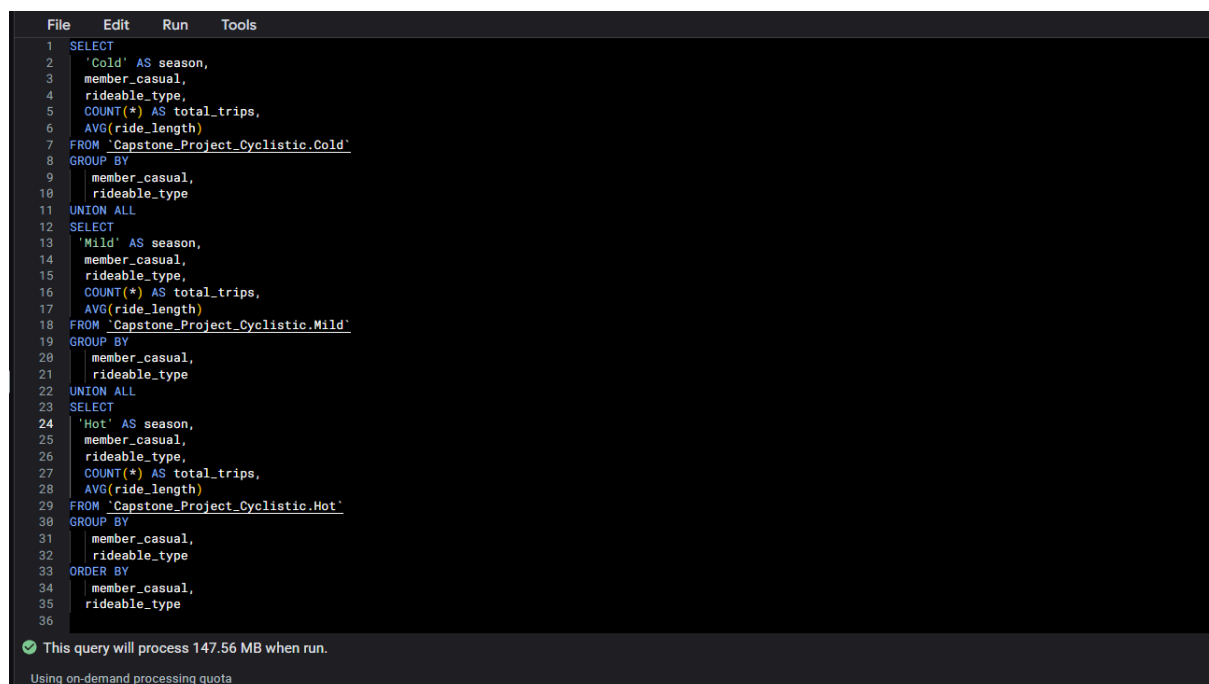
```
SELECT
    member_casual,
    rideable_type,
    COUNT(*) AS total_trips,
    AVG(ride_length)
FROM `Capstone_Project_Cyclistic.Hot`
GROUP BY
    member_casual,
    rideable_type
ORDER BY
    member_casual,
    rideable_type
```

Likewise, this code was used in order to filter the necessary data in previously made multiple temperature based tables. After getting information about all three tables, I just can not adequately analyze possible correlations due to lack of consolidated table with separate results of my formula. So right now, I am a bit confused, and in the process of thinking: What should I do next?

I had 3 choices in my thoughts:

- 1) Simply merge every table via Capcut and screenshots
- 2) Transfer data and merge my results in Google Sheets
- 3) The hardest one was to re-write the code in order to get one, which merges all my results together.

I chose the third choice because I didn't want to always use Google Sheets or editing apps for this small and simple task, so after searching in the browser for a couple of minutes I re-wrote the code into this one:



```
File Edit Run Tools
1 SELECT
2 'Cold' AS season,
3 member_casual,
4 rideable_type,
5 COUNT(*) AS total_trips,
6 AVG(ride_length)
7 FROM 'Capstone_Project_Cyclistic.Cold'
8 GROUP BY
9 member_casual,
10 rideable_type
11 UNION ALL
12 SELECT
13 'Mild' AS season,
14 member_casual,
15 rideable_type,
16 COUNT(*) AS total_trips,
17 AVG(ride_length)
18 FROM 'Capstone_Project_Cyclistic.Mild'
19 GROUP BY
20 member_casual,
21 rideable_type
22 UNION ALL
23 SELECT
24 'Hot' AS season,
25 member_casual,
26 rideable_type,
27 COUNT(*) AS total_trips,
28 AVG(ride_length)
29 FROM 'Capstone_Project_Cyclistic.Hot'
30 GROUP BY
31 member_casual,
32 rideable_type
33 ORDER BY
34 member_casual,
35 rideable_type
36
```

✓ This query will process 147.56 MB when run.
Using on-demand processing quota

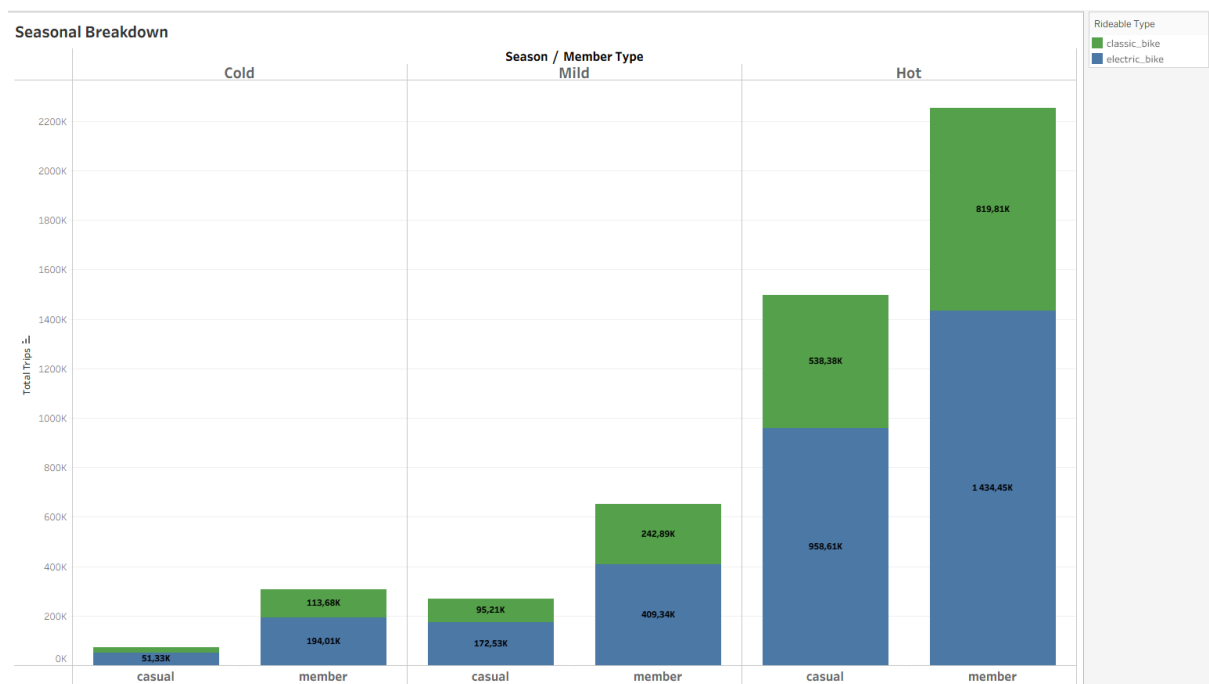
As I wasn't strongly familiar with method of using 'name' AS "type"

I firstly made a **mistake** with quotation marks and used “ ` “ instead of “ ‘ “. Because of this small mistake, I had to again surf the internet.

As a result, I got this:

Row	season	member_casual	rideable_type	total_trips	f0_
1	Hot	casual	classic_bike	538377	2474.835956959...
2	Cold	casual	classic_bike	20107	2066.265231014...
3	Mild	casual	classic_bike	95213	2277.068751115...
4	Hot	casual	electric_bike	958612	977.7356845105...
5	Cold	casual	electric_bike	51326	611.4157152320...
6	Mild	casual	electric_bike	172530	797.2052570567...
7	Hot	member	classic_bike	819807	965.8005042650...
8	Cold	member	classic_bike	113677	848.4936970539...
9	Mild	member	classic_bike	242888	911.8605077237...
10	Hot	member	electric_bike	1434447	752.5477121148...
11	Cold	member	electric_bike	194005	629.6925697791...
12	Mild	member	electric_bike	409344	679.8394870817...

However, that is still a material with which is hard to work, so let's come back to prepare phase and make it look better with Tableau



Now it looks way more clear and I can work with it way better in terms of seen patterns.

Seasonal Impact and Growth Trends:

Overall, the seasonal breakdown confirms my **hypothesis**: **higher** temperatures directly lead to **higher** bicycle usage.

Casual Users:

Casual demand grows **rapidly** as the weather warms up:

E-bikes: 51.33 k to 172.53 k to 958.61 k (multiplied by 3.36, then 5.56)

Classic bikes: 20.11 k to 95.21k to 538.38 k (multiplied by 4.73, then 5.65)

Casual users are extremely weather-dependent, and **interest** in classic bikes grows even faster than in e-bikes when it gets warm.

Members:

Member usage also **increases**, but at a much more stable rate:

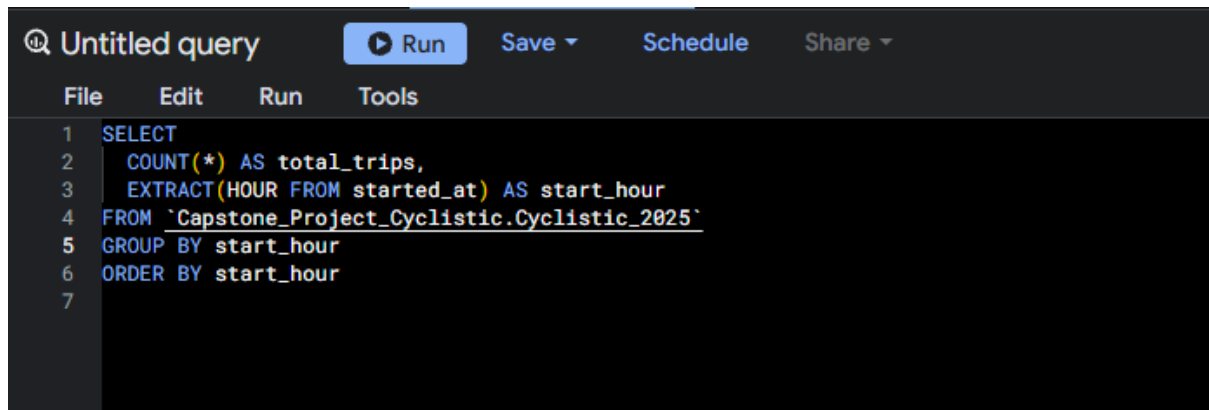
E-bikes: 194.01 k to 409.34 k to 1434.45 k (multiplied by 2.1, then 3.5)

Classic bikes: 113.68 k to 242.89 k to 819.81 k (multiplied by 2.1, then 3.37)

Conclusion:

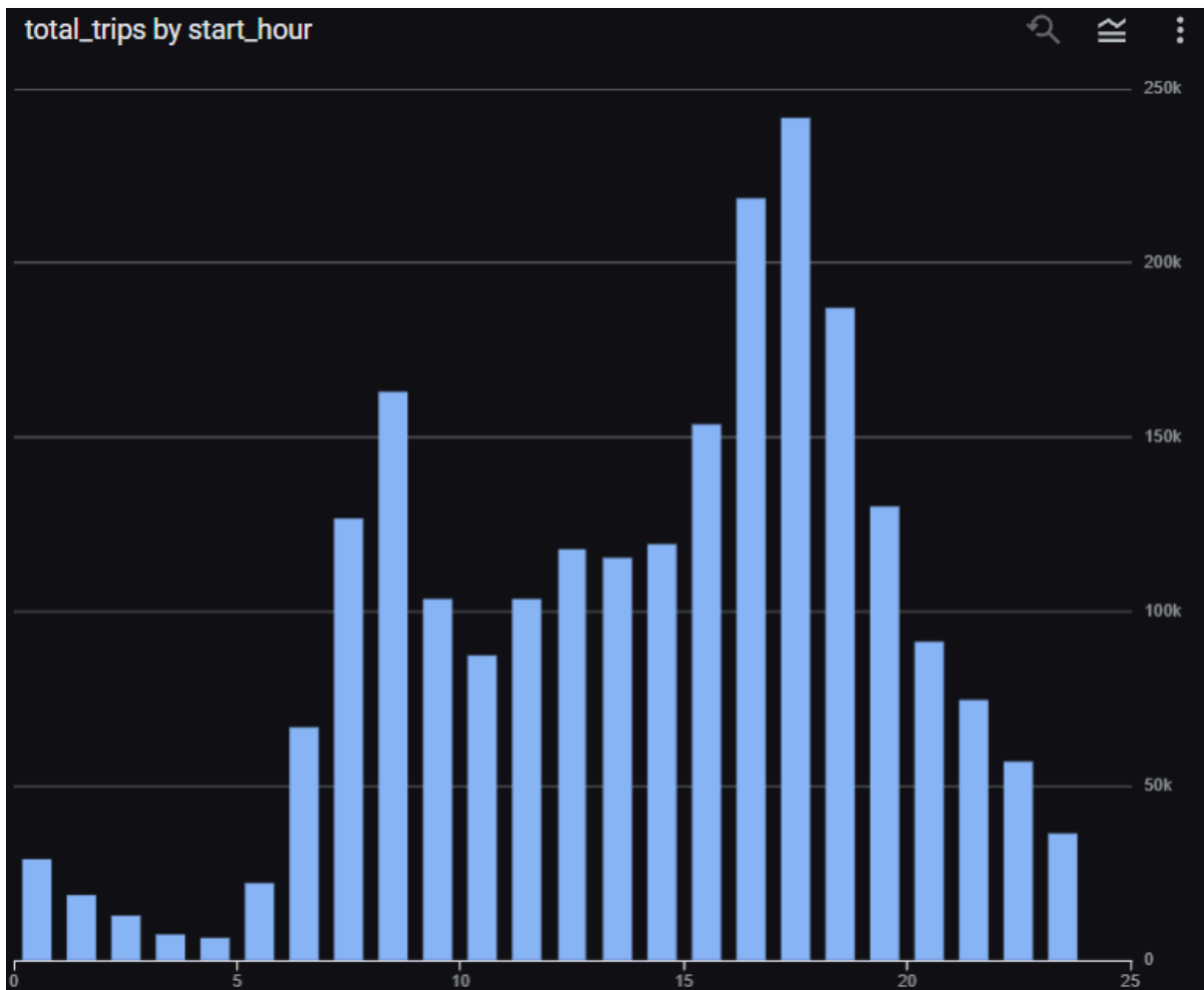
Members generate higher overall trip numbers, but casual users show much **higher** weather sensitivity. Casual growth factors average **4.05** (Cold-to-Mild) and **5.61** (Mild-to-Hot), while member growth factors stay **lower** at **2.1** and **3.44**.

The next analysis I will do in this case-study is about peak hours during a day. Here I already have a theory that the peak hours for members will be from 7 to 9 am and from 17 to 19 pm, because that is the common time for commuting to work and going home.

A screenshot of a BigQuery console interface. At the top, there's a header bar with a search icon, the text "Untitled query", and buttons for "Run", "Save", "Schedule", and "Share". Below the header is a menu bar with "File", "Edit", "Run", and "Tools". The main area is a dark-themed code editor showing a SQL query. The query is as follows:

```
1 SELECT
2   COUNT(*) AS total_trips,
3   EXTRACT(HOUR FROM started_at) AS start_hour
4 FROM `Capstone_Project_Cyclistic.Cyclistic_2025`
5 GROUP BY start_hour
6 ORDER BY start_hour
7
```

This simple code gives us overall information about total trips by hour a day. Using the button “**Visualization**” in BigQuery itself, we can create following charts



The chart shows peak hours in the morning and in the evening.

The highest amount of total trips by hour of day:

8 am – **126,55 k**

9 am – **163,12 k**

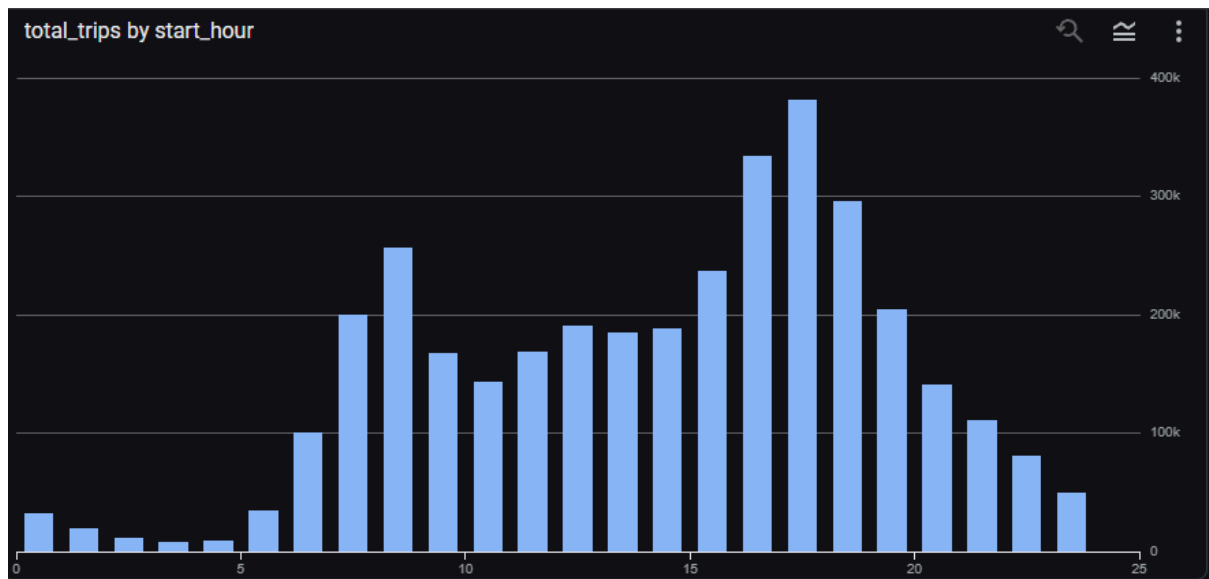
17 pm – **218,42 k**

18 pm – **241,45 k**

19 pm – **295,89 k**

Now let's get more detailed information about the type of client.

For members:



The peak hours for all members are also 8am and 9 am with 17pm and 18 pm.

The highest amount of total trips by hour of day(members):

8 am – **199,73 k**

9 am – **256,61 k**

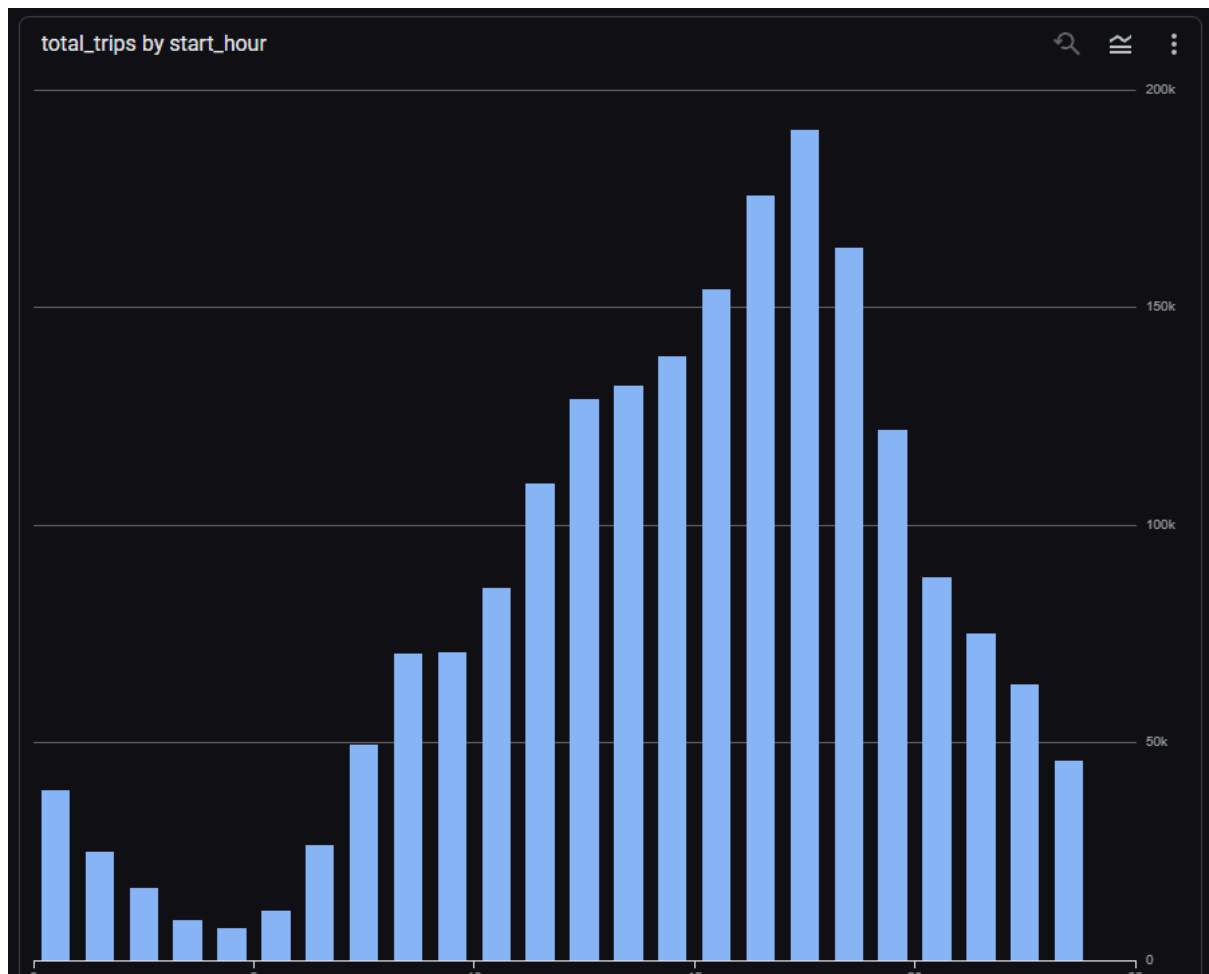
17 pm – **334,26 k**

18 pm – **381,79 k**

19 pm – **187,24 k**

This means that my theory was actually right.

For casuals:



The situation with casuals is quite different as the trend is constantly increasing starting from 6 pm at **11,44 k** straight up to 18 pm at **190,65 k**.

Hourly Usage: Daily Patterns

Members: Show two clear peak times — in the morning (8 am – 9 am, up to **256.61 k**) and in the evening (5 pm – 6 pm, peaking at **381.79 k**). This shows they mostly use bikes to travel to and from work .

Casual Users: Have a single smooth peak in the afternoon. Usage grows from 6 am (**11.44 k**) and stays highest between 12 pm and 6 pm (**109.58 k** to **190.65 k**), which means they ride mostly for fun during the day.

Key Takeaways and Findings:

Both groups prioritize electric bikes, and math says that almost twice(80% and 73%) for casuals and members, respectively. However, average ride length for casuals on classical bikes is 2.6 times longer and for members this factor is 1.3.

Casuals have trips mostly on weekends, while members peak at working days.

Members' rush hours are in the morning and in the evening, while casuals' peak hours are from 12 pm to 6 pm.

Furthermore, there is a positive correlation between temperature and the number of clients. In fact, casuals are way more temperature-sensitive than members.

Addressing Business Questions and Giving Strategic Recommendations:

1) How do annual members and casual riders use Cyclistic bikes differently?

Casuals use bikes for leisure time, while members use bikes in order to commute to work and get back home.

2) Why would casual riders buy Cyclistic annual memberships?

We should create special circumstances, so casuals would have a need for annual membership.

3) How can Cyclistic use digital media to influence casual riders to become members?

In modern times there is a trend of maintaining a healthy lifestyle. We should use this trend in our adverbs through social media with this concept:

Starting in April (last month before hot season) we launch mass advertisements in social media that show how bikes can help to lose weight, get better stamina and just help to

stay fit, and make this process beautiful through a show of riding in nature, local parks. Now, people use it not only for leisure time once or twice a week, but on a daily basis and as more and more people get membership we have to continue make them feel special. By August most of the casuals who became members for sports, should get quite familiar with bikes and here we launch a rating system in our app. People would compete with each other to be just better in their own eyes. By the end of the hot season we make an advertisement that in the last days of October there will be a race between everyone who wants to compete there.

Limitations:

This analysis has several limitations that should be considered when interpreting **the results**:

The dataset does not include rider demographics (age, gender, location of residence), which limits the ability to design precisely targeted marketing campaigns.

There is no information on the purpose of each individual trip; usage patterns (commuting vs. leisure) were inferred from time-of-day and day-of-week trends rather than stated directly.

The dataset does not track repeat casual riders, so it is not possible to identify how many casual riders already ride frequently and are strong conversion candidates versus one-time users.

Weather data was sourced separately from a general Chicago climate archive and represents average monthly temperatures rather than exact daily conditions matched to each ride, which may introduce minor inaccuracies in the seasonal correlation analysis.

Summary for people who came instantly after **ask phase**: Analysis of **5 million+** rides shows that casual riders and members use Cyclistic for **fundamentally different purposes**. Members ride primarily on **weekdays** with sharp peaks at 8–9 am and 5–6 pm, indicating commute-based usage, while casual riders peak on **weekends** and show a single broad peak between 12 pm and 6 pm, indicating leisure usage. Casual riders take significantly longer trips on average (**82% longer**) but generate **57%** fewer total rides than members. Bicycle usage for **both groups** is **strongly correlated with temperature**, with casual riders being roughly twice as weather-sensitive as members. These patterns suggest that a membership conversion strategy should focus on positioning Cyclistic as a **fitness** and **lifestyle** tool rather than a commuting alternative for casual riders.

